

On My Own

Name _____

How can you solve each equation?

1. $779 - 363 =$ _____

2. $562 - 295 =$ _____

3. $934 - 874 =$ _____

4. $134 - 68 =$ _____

5. At the airport baggage claim, there are 497 passengers and 632 pieces of luggage. How many more pieces of luggage are there than passengers?

6. Mark and Heidi are asked to solve $171 - 136$. Their work is shown. Which strategy would you choose to solve the problem?

Mark

$170 - 135 = ?$

Heidi

$136 + ? = 171$

7. Cadence earned 299 points in her online game. This was 102 points greater than her last score. Marco earned 414 points. How many more points did Marco earn?

How can you solve the equation? Explain your subtraction strategy.

8. $400 - 349 =$ _____

9. $668 - 218 =$ _____

10. On her family road trip, Zahara sees a road sign showing how many miles they are from different cities. How much farther is Fairbanks than Cantwell?



Cantwell	138
Denali Nat'l Park	166
Fairbanks	283

11. **Error Analysis** Celia says that to subtract $249 - 117$ she can decompose 249 and subtract the parts from 117. Do you agree? Explain your reasoning.

12. **Extend Your Thinking** While playing his favorite board game, Armando wins 550 points. He decides to use 335 points to buy game spaces, 122 points on a game piece, and the rest he saves. Does Armando save more than 100 points? Show how you know.

Reflect

How do you determine if a subtraction strategy is efficient?

Math is... **Mindset**

How have you worked to accomplish your goal today?

Solve Two-Step Problems Involving Addition and Subtraction



Be Curious

What question could you ask?

Lea earns points playing her favorite dance video game. She plays the next level and earns more points. Lea needs points to buy a new song for the game.



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Math is... Mindset

How can you know that you have made good decisions?

Learn

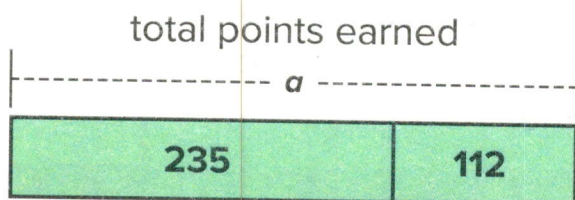
Lea earns 235 points playing her favorite dance video game. She plays the next level and earns 112 more points. Lea needs 475 points to buy a new song for the game.

How many more points does Lea need to buy a new song?

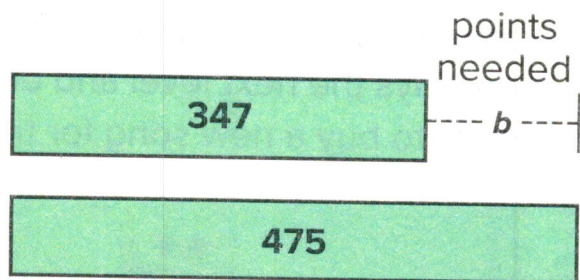
Math is... Modeling

How can you represent this problem in different ways?

Step 1 Determine how many points Lea has earned.



Step 2 Determine how many more points Lea needs.



You can use a letter to represent the unknown.

$$235 + 112 = a$$

$$235 + 112 = 347$$

Lea has 347 points.

$$475 - 347 = b$$

$$475 - 347 = 128$$

Lea needs 128 more points.

You can represent each step in a two-step problem using an equation with a letter for the unknown.

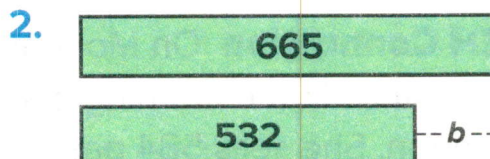
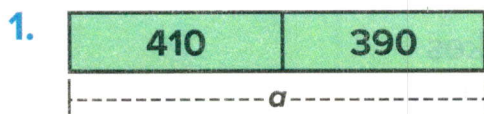
Work Together

Students collect books to donate to other schools. The first week they collect 348 books. The next week they collect 405 books. They donate 250 books to one nearby school. How can you represent the number of books they have left to donate?

On My Own

Name _____

How can you write an equation to represent the bar diagram?



How can you draw a bar diagram to represent the equation?

3. $601 - b = 299$

4. $100 + a = 750$

Represent and solve the problem. Use letters for the unknowns.

5. Sam and Ben take turns driving. They traveled 417 miles in May and 454 miles in June. If Sam drove 502 of the miles, how many miles did Ben drive?
6. Jaya earned \$187 babysitting. She bought a wireless speaker for \$129 and a carrying case for \$26. How much money does she have left?
7. Judy has 323 beads. Sarah has 142 more beads than Judy. How many beads do they have together?

How can you solve for the unknown?

8. $m = 456 - 236$
 $m = \underline{\hspace{2cm}}$

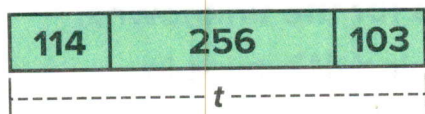
9. $867 = 235 + k$
 $\underline{\hspace{2cm}} = k$

10. **STEM Connection** On Monday, Saffron bakes 324 pastries in the morning and some more in the afternoon. She sells 584 pastries. At the end of the day she has 52 pastries left. How many pastries did she bake in the afternoon?



11. Elroy's Balloon Emporium sells boxes of balloons. There are 100 balloons in a box. In January, they sell 3 boxes. In February, they sell 6 boxes. If they started with 10 boxes, how many balloons do they have left to sell?

12. **Extend Your Thinking** The bar diagram represents a two-step word problem. Write a two-step word problem that could represent the bar diagram.



Reflect

How can you make sense of a two-step problem?

Math is... Mindset

How do you know that you made good decisions?

Compare 4-Digit Numbers



Be Curious

What do you notice?
What do you wonder?



Math is... Mindset

How can a different perspective help you with your work today?

Learn

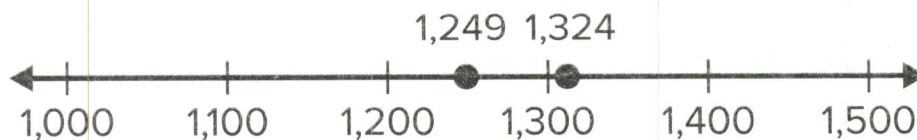
James and Phoebe both collect pennies. James has 1,324 pennies and Phoebe has 1,249.

Who has more pennies?

You can use different methods to compare numbers.



► **One Way** Use a number line. Plot 1,324 and 1,249 on the number line.



1,324 is to the right of 1,249 on the number line. So, $1,249 < 1,324$.

► **Another Way** Use place value.

Compare the thousands digits first.

1,324
1,249

If the thousands digits are the same, compare the hundreds digits.

3 hundreds is not equal to 2 hundreds, so $1,324 \neq 1,249$.

3 hundreds is greater than 2 hundreds, so $1,324 > 1,249$.

James has more pennies than Phoebe.

You can use number lines or place value to compare 4-digit numbers.

Math is... Verifying

How can you verify the numbers are compared correctly?

Work Together

Which is greater? Write $<$, $>$, or $=$ to compare the numbers. Explain your thinking.

4,525 ○ 4,523

On My Own

Name _____

How can you compare the numbers? Plot each number on the number line, then write $<$, $>$, or $=$.

1. 3,750 2,000



2. 4,402 4,516



Are the numbers equal or not equal? Write $=$ or $\neq$.
Then explain your thinking.

3. 8,429 8,429

4. 7,110 7,129

Which is greater? Write $<$ or $>$.
Then explain your thinking.

5. 3,345 2,740

6. 1,320 1,732

How can you compare the numbers? Write $<$, $>$, or $=$.

7. 4,349 349

8. 2,248 2,890

9. 3,877 3,874

10. 945 4,739

11. **STEM Connection** Poppy is organizing supplies. She wants to compare sizes of two park field houses. One building holds 2,389 people. The other holds 1,288 people. Which holds more people?



12. **Extend Your Thinking** How can you compare 3,489, 2,889, and 3,288?

Reflect

What did you notice about comparing 4-digit numbers?

Math is... Mindset

How did different perspectives help you with your work today?

Fluently Add Multi-Digit Numbers



Be Curious

What question could you ask?

Last year the library's collection of books increased.
This year additional books were added to the library.

Math is... Mindset

How can your skills or interests help you with your work today?

Learn

Last year the library's collection of books grew from 1,350 to 2,322 books. This year 1,569 books were added to the library.

How can you find the total number of books in the library?

Math is... Analyzing

How can you decide which information in the problem is useful?

Decompose an Addend

Decompose one addend by place value and add the parts.

$$2,322 + 1,569 = ?$$

$$2,322 + 1,000 = 3,322$$

$$3,322 + 500 = 3,822$$

$$3,822 + 60 = 3,882$$

$$3,882 + 9 = 3,891$$

$$2,322 + 1,569 = 3,891$$

Partial Sums Add by place value to find partial sums. Then add the partial sums to find the total.

$$\begin{array}{r} 2,322 \\ + 1,569 \\ \hline 3,000 \\ 800 \\ 80 \\ + 11 \\ \hline 3,891 \end{array}$$

Use an Algorithm Add by place value starting with the ones place and moving left.

$$\begin{array}{r} 2,322 \\ + 1,569 \\ \hline 3,891 \end{array}$$

Regroup 11 as 1 ten and 1 one.

You can use any addition strategy to find the sum. You can decide which addition strategy is most efficient for you.

Work Together

Angel and Daniel find the sum of $4,348 + 6,227$. Angel uses an addition algorithm. Daniel uses partial sums. Can either strategy be used to solve? Explain your reasoning.

On My Own

Name		
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How can you solve the equation?

1. $437 + 269 =$

2. $= 123 + 2,443$

3. $= 3,672 + 4,562$

4. $17,016 + 1,111 =$

5. Error Analysis Priya is adding $1,125 + 478$. Priya starts by adding 75 to 1,125. Then she adds 400. Then she adds 3. Do you agree with her strategy? Explain.

6. Sam adds $529 + 286$. His work is shown. What is another way Sam could find the sum? Show what his work would look like.

$$\begin{array}{r} 529 \\ + 286 \\ \hline 700 \\ 100 \\ + 15 \\ \hline 815 \end{array}$$

7. Marcus walked 2,074 fewer steps than his sister Tanya. If Marcus walked 7,488 steps, how many steps did his sister walk?

How can you solve the equation? Explain your addition strategy.

8. $719 + 234 =$ _____

9. $9,248 + 132 =$ _____

10. Mr. Perez needs 650 sheets of construction paper for his class art project. How many packs of this construction paper should he buy?



11. Tryon School raised \$2,345 at their school fundraiser and \$1,886 at their bake sale. They used \$650 of the money they raised to buy new playground equipment. How much money did Tryon School raise?
12. **Extend Your Thinking** The community center had 113 swimmers in the fall class and 235 swimmers in the winter class. The summer class had 105 more swimmers than both classes combined. How many swimmers were in the summer class?

Reflect

How do you decide which addition strategy to use when solving a problem?

Math is... Mindset

How did you use your skills or interests to help you with your work?

Use an Algorithm to Subtract



Be Curious

What question could you ask?



African Elephant
8,386 pounds



Asian Elephant
7,185 pounds

Math is... Mindset

What goal do you want to accomplish today?

Learn

How can you determine the difference in weight between the African elephant and the Asian elephant?



African Elephant
8,386 pounds



Asian Elephant
7,185 pounds

You can use an algorithm to subtract. Subtract the digits by place value. Start with the ones place and move to the left.

Step 1 Subtract the ones.
 $6 - 5 = 1$

Step 2 Subtract the tens.
 $80 - 80 = 0$

$$\begin{array}{r} 8,386 \\ - 7,185 \\ \hline 01 \end{array}$$

Step 3 Subtract the hundreds.
 $300 - 100 = 200$

Step 4 Subtract the thousands.
 $8,000 - 7,000 = 1,000$

$$\begin{array}{r} 8,386 \\ - 7,185 \\ \hline 1,201 \end{array}$$

The African elephant weighs 1,201 pounds more than the Asian elephant.

Math is... Structure

How can you apply what you know about the addition algorithm to use the subtraction algorithm?

An algorithm can be a more efficient way of subtracting numbers.

Work Together

A school play sold 6,728 tickets for two shows. They sold 4,203 tickets for the first show. How many tickets did they sell for the second show? Use an algorithm to solve.

On My Own



Name _____

What is the difference? Use an algorithm to solve.

1.
$$\begin{array}{r} 558 \\ - 247 \\ \hline \end{array}$$

2.
$$\begin{array}{r} 975 \\ - 462 \\ \hline \end{array}$$

3.
$$\begin{array}{r} 4,964 \\ - 803 \\ \hline \end{array}$$

4.
$$\begin{array}{r} 8,579 \\ - 422 \\ \hline \end{array}$$

5.
$$\begin{array}{r} 2,646 \\ - 1,335 \\ \hline \end{array}$$

6.
$$\begin{array}{r} 9,823 \\ - 2,301 \\ \hline \end{array}$$

7.
$$\begin{array}{r} 7,438 \\ - 5,225 \\ \hline \end{array}$$

8.
$$\begin{array}{r} 17,982 \\ - 2,580 \\ \hline \end{array}$$

-
9. Addie and her family are driving to Florida to see her grandmother. The trip is 1,387 miles. They drove 365 miles the first day. How many miles do they have left to drive?

10. A summer camp is building new cabins. They spent \$2,789 for wood and tools to complete the project. The tools cost \$1,024. How much did the summer camp spend on wood?

11. Fill in the missing digits. Explain how you found each digit.

$$\begin{array}{r} 2,74\Box \\ - 1,\Box\Box4 \\ \hline \Box,541 \end{array}$$

12. **Extend Your Thinking** Starwood School has an annual walkathon. Was the increase greater from two years ago to last year, or from last year to this year? Explain how you know.

Walkathon Fundraiser		
2 Years Ago	Last Year	This Year
\$3,420	\$5,652	\$7,875

Reflect

Why is it helpful to write numbers vertically when you use a subtraction algorithm?

Math is... Mindset

How have you worked to accomplish your goal today?

Unit Review

Name _____

Vocabulary Review

Choose the correct word to complete each sentence.

compatible number

decompose

estimate

expanded form

round

1. When you _____ addends, you break them apart.
(Lesson 2-7)
2. You can _____ an answer when you do not need an exact answer. (Lesson 2-3)
3. When you _____ a number, you find a number that is close to the original number. (Lesson 2-2)
4. A _____ is a number that is close to a number but is easy to work with. (Lesson 2-3)
5. The representation of a number as a sum of the values of each digit is called _____. (Lesson 2-1)

Review

6. Which is the standard form of three thousand, eight hundred forty? (Lesson 2-1)

A. 384
B. 3,408
C. 3,804
D. 3,840

7. Circle the choice that correctly shows rounding to the nearest 10. (Lesson 2-2)

784 → 785
563 → 560
492 → 500
535 → 530

8. How can you solve the equation? Explain your addition strategy. (Lesson 2-14)
- $5,275 + 4,436 =$ _____

9. What is the difference?

(Lesson 2-15)

$$\begin{array}{r} 67,249 \\ - 24,034 \\ \hline \end{array}$$

Complete the following using properties of addition. (Lesson 2-4)

10. $289 + 621 = 621 +$ _____

11. Sasha spent 284 minutes swimming the first week and 247 minutes the second week. What is an estimate of the total time Sasha spent swimming? Explain your reasoning. (Lesson 2-3)

Will the sum be even or odd?

Write *even* or *odd*. (Lesson 2-5)

12. The sum of two numbers that are both odd will always be _____.

13. Use partial sums to help find the sum. (Lesson 2-6)

$506 + 472 =$ _____

- 14.** Decompose one number to help find the difference.

(Lesson 2-7)

$$367 - 228 = \underline{\hspace{2cm}}$$

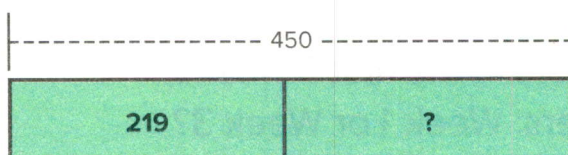
- 15.** Gabe is trying to solve $246 + 367$. Which equation shows how he could adjust the addends to find the sum?

(Lesson 2-8)

- A.** $250 + 360 = ?$
- B.** $250 + 370 = ?$
- C.** $250 + 363 = ?$
- D.** $250 + 371 = ?$

- 16.** Use the bar diagram to help you find the missing number.

(Lesson 2-9)



$$219 + \underline{\hspace{2cm}} = 450$$

$$450 - \underline{\hspace{2cm}} = 219$$

- 17.** Do you think it makes more sense to decompose both addends or adjust both addends to solve $388 + 472$? Explain. (Lesson 2-10)

- 18.** Which of the following equations can you use to solve this subtraction equation?

(Lesson 2-11)

$$836 - 377 = ?$$

- A.** $377 + 459 = 836$
- B.** $459 - 377 = 836$
- C.** $459 - 836 = 377$
- D.** $836 + 459 = 377$

- 19.** Enrique read 249 pages of his book in June and 227 pages of his book in July. The book has a total of 638 pages and Enrique wants to know how many pages he has left to read. (Lesson 2-12)

Which set of equations could be used to solve the problem?

- A.** $227 + 249 = 476$
 $638 - 227 = 411$
- B.** $249 + 227 = 476$
 $638 - 476 = 162$
- C.** $638 - 249 = 389$
 $389 + 227 = 616$
- D.** $638 - 227 = 411$
 $411 + 249 = 660$

- 20.** Complete the comparison using $>$, $<$, or $=$. (Lesson 2-3)

$$7,841 \bigcirc 1,685$$

Performance Task

Saffron is helping in a new pastry shop that has been open for two weeks.

Pastry Shop Customers	
Week	Number of Customers
1	324
2	289
3	?

Part A How many total customers did the pastry shop have in Weeks 1 and 2?

Part B What is the difference between the number of customers in Weeks 1 and 2?

Part C The pastry shop's goal is to have 800 customers within the first 3 weeks. How many customers must visit in Week 3 for the shop to reach the goal?

Part D If the pastry shop has 847 visitors in the first 3 weeks, which week would have more customers: Week 1 or Week 3? How many more?

Reflect

How can you use strategies to add and subtract multi-digit numbers?

Unit 2

Fluency Practice

Name _____

Fluency Strategy

You can use a 10 to add or subtract.

Make a 10 to add $49 + 22$.

49 is 1 away from a ten.

Subtract 1 from 22 and add 1 to 49.

$$\begin{array}{r} 49 + 22 = ? \\ \begin{array}{c} +1 \\ \downarrow \end{array} \quad \begin{array}{c} -1 \\ \downarrow \end{array} \\ 50 + 21 = 71 \end{array}$$

Use a 10 to subtract $75 - 43$.

43 is 3 away from a ten.

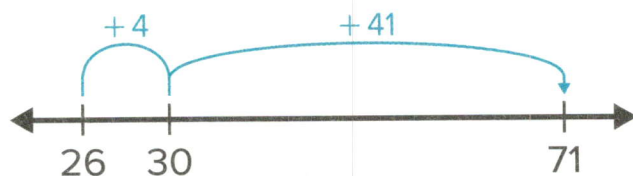
Subtract 3 from 75 and subtract 3 from 40.

$$\begin{array}{r} 75 - 43 = ? \\ \begin{array}{c} -3 \\ \downarrow \end{array} \quad \begin{array}{c} -3 \\ \downarrow \end{array} \\ 72 - 40 = 32 \end{array}$$

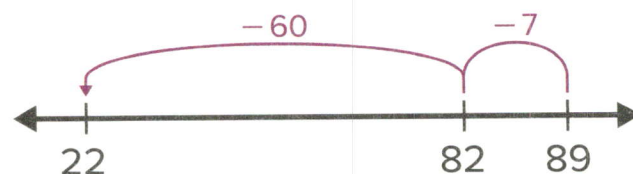
Fluency Flash

How can you represent the number line with an equation?

1.



2.



Fluency Check

How can you find the sum or difference?

3. $49 + 16 =$ _____

4. $75 + 0 =$ _____

5. $76 - 42 =$ _____

6. $58 + 37 =$ _____

7. $87 - 2 =$ _____

8. $29 - 1 =$ _____

9. $61 + 0 =$ _____

10. $58 + 2 =$ _____

11. $89 - 23 =$ _____

12. $19 + 2 =$ _____

13. $52 - 1 =$ _____

14. $54 - 31 =$ _____

15. $17 + 25 =$ _____

16. $49 + 48 =$ _____

Fluency Talk

How can you explain how to make 10 to add $28 + 17$?

How is adjusting to make a 10 different when subtracting than when adding?